Computing research

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Adapted from previous slides of Dr. Ethel Ong

What is research?

Research

Oxford: The systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions

Merriam-Webster: 1. careful or diligent search; 2. studious inquiry or examination; especially : investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws; 3. the collecting of information about a particular subject

Why research?

Creation of new knowledge/Improves existing knowledge

Improves quality of life

Advances society

Improves our understanding of the world

Research process

Problem identification

Solution design

Evaluation

Communication

Skills from doing research

New knowledge
Independent, critical thinking skills
Improved communication skills
Importance of providing supporting evidence

Qualities of good researchers

- Curious
- Resilient to setbacks
- Thick-skinned
- Open-minded
- Courageous
- Communicative, articulate, good at selling
- Self motivated, hard working, driven, committed

What is computer science?

Computer science

"the study of computers and the major phenomena that surround them," – *Newell, et al (1967)*

"the systematic study of algorithmic processes that describe and transform information: their theory, analysis, design, efficiency, implementation, and application" – *Denning, et al (1988)*

Newell, A., Perlis, A. & Simon, H. (1967). What is computer science? Science, 157, 1373-4.
Denning, P. J., et al. (1988). Report on the ACM Task Force on the Core of Computer Science. New York: ACM Press (Order No. 201880).

Computer science

"Computer science (CS) is the study of computers and algorithmic processes, including their principles, their hardware and software designs, their applications, and their impact on society." – *Tucker, et al (2003)*

Tucker, Allen, (editor), Deek, F., Jones, J., McCowan, D., Stephenson, C., and Verno, A. , A Model Curriculum for K-12 Computer Science: Final Report of the ACM K-12 Task Force Curriculum Committee. Association for Computing Machinery (ACM), New York, New York, October, 2003. (2nd ed., 2006.)

Research Areas

- artificial intelligence
- computer systems and networks
- security
- database systems
- human-computer interaction
- vision and graphics
- numerical analysis
- programming languages
- software engineering
- bioinformatics
- theory of computing

What is computer science? https://undergrad.cs.umd.edu/what-computer-science

Where to look?

- First-party: run by academic associations (IEEE, ACM) or publishers themselves
  - ACM Digital Library (https://dl.acm.org)
  - IEEE Xplore Digital Library (https://ieeexplore.ieee.org/Xplore/home.jsp)
  - Elsevier ScienceDirect (https://www.sciencedirect.com)
  - Springer Link (https://link.springer.com)
- Third-party: run by external organisations
  - Google Scholar (https://scholar.google.com)
  - ResearchGate (https://www.researchgate.net)
  - Allen Institute's Semantic Scholar (https://www.semanticscholar.org)

Questions?